United International University
Department of Computer Science and Engineering
CSE 2217: Data Structure and Algorithms II
Final Examination: Spring 2026, Duration: 2 Hours, Total Marks: 40
Answer all questions. Marks are indicated on the right side of each question.

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

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1. a) The current state of a Disjoint Set Union (DSU) data structure is represented by the following arrays.

Index	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Parent	1	3	3	3	5	7	7	7	9	9
Rank	0	0	0	2	0	0	0	2	0	1

Assume that Union-by-Rank and Path Compression are used.

i. Draw the corresponding disjoint-set forest. [1]

ii. Execute Find-Set(4). Draw the Parent array caused by path compression (if any). [1]

iii. Perform Union(2, 6). Draw the updated Parent array and Rank updates. [2]

iv. After all operations, list the representative of each disjoint set and state the total number of disjoint sets. [1]

b) Assume the following DSU trees exist before a UNION operation:

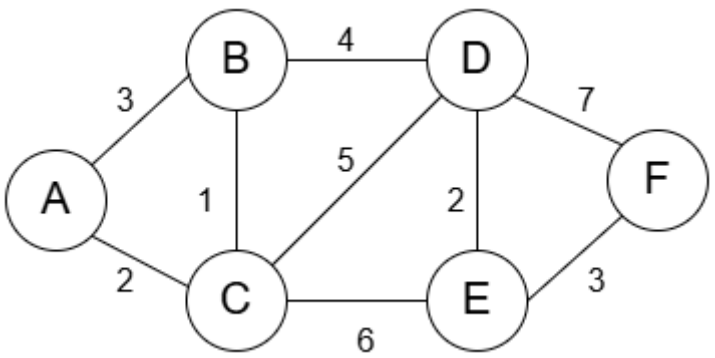
- Root A has rank 3
- Root B has rank 2
- Root C has rank 3

Answer the following:

i. If UNION(A,B) is performed using Union by Rank, which node becomes the parent and why? [1]

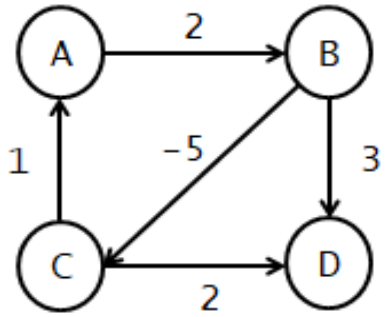
ii. If UNION(A,C) is performed, what will be the rank of the resulting root after the union? Explain. [1]

2. a) Find the Minimum Spanning Tree (MST) using Prim's algorithm for the following graph (starting vertex: A). Draw the final MST and also calculate the cost. You must show your working. [5]



b) Can a connected weighted graph have more than one MST? Justify your answer and give an example. [2]

3. Consider the following graph



a) Does there exist a shortest path from A to D in this graph? Explain with proper reasoning. [2]

b) Which algorithm will help you decide whether such a shortest path exists? [1]

c) Simulate the chosen algorithm step-by-step and show how it reaches its final conclusion regarding the path from A to D. [4]

4. Consider an open-addressing hash table of size 11. Collisions are resolved using quadratic probing with the hash function:

$$h(k,\ i)\ =\ (h'(k)\ +\ 3\,i^2)\ mod\ 11;\ where\ h'(k)\ =\ (2k\ +\ 5)\ mod\ 11$$

The table currently contains the following entries:

0	1	2	3	4	5	6	7	8	9	10
14	NIL	NIL	NIL	60	33	NIL	34	NIL	NIL	NIL

By showing detailed hash value calculations at each step, redraw the table after each of the following operations performed in sequence:

i) Insert key 46. [1]

ii) Insert key 82. [2]

iii) Delete key 34 from the table by replacing its slot with NIL. [1]

iv) If your search of 82 fails despite being present at the table, what might be the possible reason? Explain how you can modify the delete operation to prevent this from happening. [2]

v) State one merit and one demerit of quadratic probing as a collision resolution strategy. [1]

5. A football analytics platform used during FIFA World Cup broadcasts applies a modified Rabin–Karp string-matching algorithm with a simple position-independent block hash to scan live commentary transcripts for key terms (e.g., "TITLE" as in title race or title decider).

Text (T): **SAVESTITLE**

Pattern (P): **TITLE**

Character Values (A = 1, B = 2, C = 3, … Z = 26):

Char	S	A	V	E	T	I	L
Value	19	1	22	5	20	9	12

Hash Function: H(s) = (C1+C2+C3+C4+C5)

Where: C1, C2, C3, C4, C5 are the numeric values of the characters in the substring.

Rolling Hash Update Rule:

$H_{new} = (H_{old} - outgoing\ character\ value + incoming\ character\ value)$

a) Compute the hash value of the pattern "TITLE". Show your calculation. [2]

b) Compute hash values for all length-5 substrings of the text "SAVESTITLE" using the rolling hash technique. Identify the index of the Valid Hit & Spurious Hits. (Clearly show window-wise hash calculations using rolling updates). [4]

c) Determine the worst-case time complexity of the modified Rabin–Karp algorithm using the given hash function. Briefly justify your answer. [1]

6. A logistics company has developed an AI-based system to plan delivery routes across multiple cities. The system must determine whether there exists a valid route that:

- Starts from a source city,
- Visits all required cities at least once,
- Returns to the starting city,
- And keeps the total travel cost within a given fuel budget.

To solve this, the system explores many possible routes in a highly parallel way and attempts to identify a valid one if it exists. However, while it is easy to verify whether a given route satisfies all conditions, it is not always efficient to determine whether such a route exists when the number of cities becomes large.

As the problem size grows, the computation becomes increasingly complex, and the outcome cannot be determined efficiently in all cases.

a) In which class does this problem belong: P or NP? Explain your answer. [2]

b) What does it mean if this problem is proven to be NP-Complete? [1]

c) Explain the relationship among P, NP, NP-Complete, and NP-Hard classes using a diagram. [2]